

1. Declare a function called `getUserInitials` that takes a user's full name (like 'Virat Kohli') and returns their initials in uppercase (e.g., 'VK'). Call this function with your favorite cricketer's name and print the result.

```
2. #include <stdio.h>
3. #include <ctype.h>
4.
5. void getUserInitials(char name[]) {
6.     printf("Initials: ");
7.
8.     // Print the first character
9.     printf("%c", toupper(name[0]));
10.
11.    // Print the first letter after every space
12.    for (int i = 1; name[i] != '\0'; i++) {
13.        if (name[i - 1] == ' ') {
14.            printf("%c", toupper(name[i]));
15.        }
16.    }
17.
18.    printf("\n");
19.}
20.
21.int main() {
22.    char name[] = "Virat Kohli";
23.
24.    getUserInitials(name);
25.
26.    return 0;
27.}
```

2. Create a function `addToCart` that takes a shopping cart array and a product name, adds the product to the cart, and prints the updated cart. Demonstrate how passing the cart array by reference allows changes to persist outside the function.

Hint:
In languages like JavaScript, arrays are passed by reference. In C/C++, use pointers for reference behavior.

```
#include <stdio.h>
#include <string.h>

void addToCart(char cart[][20], int *count, char product[]) {
    strcpy(cart[*count], product); // Add product to cart
    (*count)++;                     // Increase item count

    printf("\nUpdated Cart:\n");
    for (int i = 0; i < *count; i++) {
        printf("%d. %s\n", i + 1, cart[i]);
    }
}
```

```

int main() {
    char cart[10][20] = {"Laptop", "Mouse"};
    int count = 2;

    printf("Cart Before Adding:\n");
    for (int i = 0; i < count; i++) {
        printf("%d. %s\n", i + 1, cart[i]);
    }

    addToCart(cart, &count, "Keyboard");

    printf("\nCart After Function Call:\n");
    for (int i = 0; i < count; i++) {
        printf("%d. %s\n", i + 1, cart[i]);
    }

    return 0;
}

```

3 .Write two functions: `increaseFollowersByValue` and `increaseFollowersByReference`. Each should take a followers count and add 1000 to it, but one should use pass-by-value and the other pass-by-reference. Show how the original followers count changes (or doesn't) after calling each function.

```

#include <stdio.h>

// Pass by Value
void increaseFollowersByValue(int followers)
{
    followers = followers + 1000;
    printf("Followers inside increaseFollowersByValue: %d\n", followers);
}

// Pass by Reference
void increaseFollowersByReference(int *followers)
{
    *followers = *followers + 1000;
    printf("Followers inside increaseFollowersByReference: %d\n", *followers);
}

int main()
{
    int followers = 5000;

    printf("Original Followers: %d\n", followers);

    // Pass by Value

```

```

    increaseFollowersByValue(followers);
    printf("Followers after increaseFollowersByValue: %d\n", followers);

    // Pass by Reference
    increaseFollowersByReference(&followers);
    printf("Followers after increaseFollowersByReference: %d\n", followers);

    return 0;
}

```

4. Build a function `formatPrice` that takes a price in rupees and returns a string formatted like Flipkart's price tag (e.g., '₹1,599'). Use this function to display prices for three different products.

```

#include <stdio.h>
#include <string.h>

char* formatPrice(int price) {
    static char formatted[20];
    char temp[20];
    int i, j = 0, count = 0;

    sprintf(temp, "%d", price);

    for (i = strlen(temp) - 1; i >= 0; i--) {
        formatted[j++] = temp[i];
        count++;

        if (count == 3 && i != 0) {
            formatted[j++] = ',';
            count = 0;
        }
    }

    formatted[j] = '\0';

    // Reverse the string
    int start = 0, end = j - 1;
    while (start < end) {
        char ch = formatted[start];
        formatted[start] = formatted[end];
        formatted[end] = ch;
        start++;
        end--;
    }

    static char result[25];
    sprintf(result, "₹%s", formatted);

    return result;
}

```

```

}

int main() {
    int laptop = 1599;
    int mobile = 25999;
    int headphones = 3499;

    printf("Laptop      : %s\n", formatPrice(laptop));
    printf("Mobile      : %s\n", formatPrice(mobile));
    printf("Headphones   : %s\n", formatPrice(headphones));

    return 0;
}

```

5. Refactor an existing function you wrote above to make it reusable for both product names and usernames (for example, a function that capitalizes the first letter of any string).

Constraint: The refactored function should work for any string input, not just a specific use case.

```

#include <stdio.h>
#include <ctype.h>

// Reusable function
void capitalizeFirstLetter(char str[]) {
    if (str[0] != '\0') {
        str[0] = toupper(str[0]);
    }
}

int main() {
    char productName[] = "laptop";
    char userName[] = "ridham";

    capitalizeFirstLetter(productName);
    capitalizeFirstLetter(userName);

    printf("Product Name: %s\n", productName);
    printf("Username    : %s\n", userName);

    return 0;
}

```